An aerial image is worth more than a single class: Model limitations for class and image imbalance in multilabel classification.

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Introduction

The deep learning domain has emerged as a transformative paradigm during the last decade being adopted in multiple research fields such as climate studies, medicine, or agriculture (Kamilaris & Prenafeta-Boldú, 2018; LeCun et al., 2015). This ongoing urge comes from the increase in data availability where the volume has exploded globally with public datasets, specific pipelines of work, and cheap storage capacity. Hardware capacity is another important part where the GPU power allows it to scale into more complex and bigger models. All these aspects combined with important software development such as open code libraries and pretrained models, make deep learning more accessible to everyone.

Remote sensing is one of the domains that has adopted deep learning to solve complex tasks such as: image classification, object detection, and time series analysis among others (Ma et al., 2019). In this project a multilabel image classification problem will be evaluated using the UC-Merced archive from (Chaudhuri et al., 2018), which consists of 2,100 high-resolution aerial RGB images of 256×256 pixels and a spatial resolution of 0.3 m per pixel, extracted from the USGS National Map Urban Area Imagery collection. Although the dataset also provides single-label annotations with 21 classes, we will focus exclusively on the multi-label classification task, given its greater complexity and closer resemblance to real-world heterogeneous land surfaces, where multiple objects typically coexist within a single image (Karalas et al., 2015).

The general objective of this project is to evaluate the performance of 2 different pretrained DL architectures (ResNet and ViT) under conditions of class imbalance, specifically samples-per-class and classes-per-image imbalance. Under this scenario, our research questions are:

Research questions:

1) How do ResNet50 and ViT-B/16 models compare in multilabel classification performance on the UC Merced dataset under samples-per-class imbalance, using consistent hyperparameter settings?

2) How do ResNet50 and ViT-B/16 models perform in classifying UC Merced Multilabel *classes per image* imbalance considering default model parameters?

We hypothesize that; ***1) pre-trained models can generalize effectively despite class imbalance, regardless of whether it manifests as samples-per-class or classes-per-image imbalance, and 2) pretrained models on single image classification will perform poorly at increasing multilabel classification.***

Methods

In order to keep maximum control over the model comparison, initial data preparation is essential. For this reason the workflow was divided in 2 stages, the first one consisted in data preparation while the second part corresponds to the model selection, tuning and assessment.

1) Data preparation:

Although the UC-Merced dataset is originally organized into 21 mutually exclusive land-use categories — one folder per class, with each image assigned to exactly one category — this work reframes the task as a multi-label classification problem using the accompanying binary annotation file. This file maps each image to 17 semantic attributes (e.g., buildings, trees, water, bare-soil) that describe the visual content of the scene independently, allowing multiple attributes to be simultaneously present in a single image. The shift from 21 single-label classes to 17 multi-label attributes reflects a change in the nature of the target variable: rather than predicting a mutually exclusive land-use category, the model must produce a binary prediction for each semantic concept. Because the *samples per class* and *classes per image* are imbalanced, a naive random split risks leaving rare classes or rare images under-represented or absent from the validation and test partitions. To address this, a multi-label stratified shuffle split was applied, which preserves the marginal class

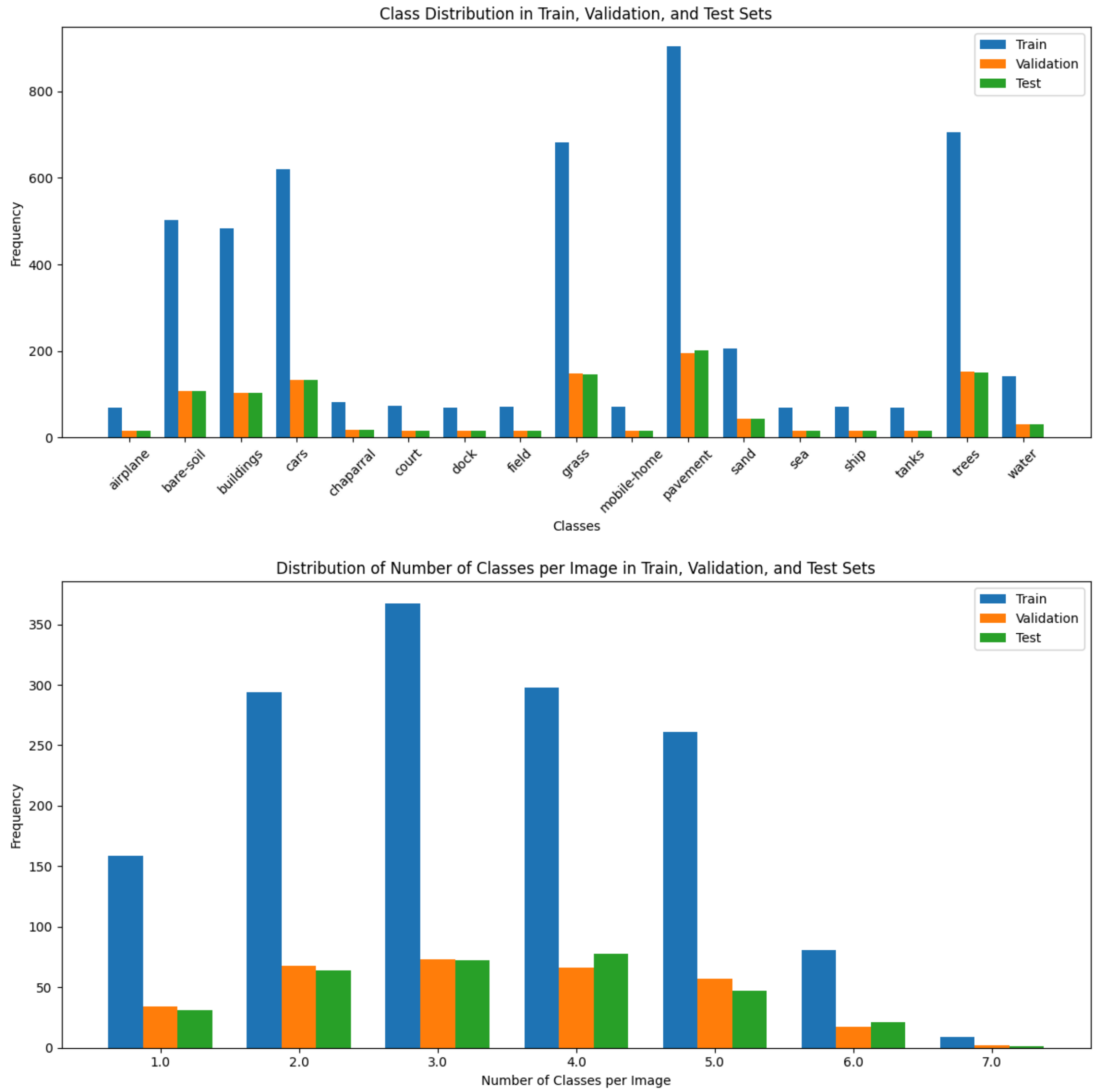


Figure 1: Samples per class and classes per image imbalance and distribution in train, test, validation datasets

and image distribution across all three subsets (Sechidis et al., 2011). The data was divided into 70% training, 15% validation, and 15% test in two sequential seeded stages to avoid data leakage between sets and ensure reproducibility. This is shown in Figure 1.

For data augmentation and normalization, the training images are resized to 224×224 pixels and augmented with random horizontal and vertical flips, rotations up to 45°, and color jitter over brightness, contrast, and saturation. Validation and test images are only resized, with no augmentation applied. All partitions are normalized using ImageNet channel statistics to match the pretraining distribution of both backbones. Finally, the three partitions are loaded into iterators with a shared batch size and configuration, where only the training iterator is shuffled to ensure reproducible evaluation.

2) Tested models:

Why use pretrained models

This transfer learning strategy is particularly justified for UC-Merced (2,100 images) where the convolutional backbone already encodes a rich hierarchy of visual representations — from low-level edges to high-level semantic patterns reducing the effective data requirement for the downstream multi-label task. We implement ResNet and ViT architectures pretrained on the ImageNet database with 1K classes because it was used as benchmark in both Residual Network and Vision Transformer publications (He et alt, 2016; Dosovitskiy et al., 2021). This means the findings and conclusions from these papers refer directly to the model we implement and thus can help in discussion.

Why use ResNet and ViT architectures

The breakthroughs in computer vision caused by Residual network and Vision transformer architectures have been the most influential in the last decade. Comparing these shows us strengths and weaknesses in model architecture which is the primary purpose of this research. We implement "smaller" versions of these architectures, particularly ResNet50, referring to a 50-layer residual network architecture, and a ViTB16 referring to the Base model with 12 transformer block layers, 768 hidden layers, and 12 self-attention heads (Devlin et al., 2018). "16" refers to the input patch size, dividing the image in 16x16 patches fed to the transformer (Dosovitskiy et al., 2021). This model choice is based on the computational constraints and marginal improvements to performance to deeper versions.

Why BCElogitSLoss as loss function

The loss function used was BCEWithLogitsLoss (binary cross-entropy loss), which corresponds to the standard cross-entropy loss with sigmoid activation widely adopted as one of the baseline for multi-label classification in remote sensing (Yessou et al., 2020). Unlike softmax-based losses that treat classes as mutually exclusive, it decomposes the problem into C independent binary classifications — one per class — computing a separate sigmoid and binary cross-entropy for each logit. This allows multiple classes to be simultaneously active for a single image, which is the core requirement of the UC-Merced multi-label task. A pos_weight vector was additionally applied to compensate for class imbalance by up-weighting underrepresented labels during training.

Why do we use identical Learning rates and Weight decays?

Both ViT-B/16 and ResNet-50 were fine-tuned under the same optimization regime to ensure a controlled comparison in which any performance difference can be attributed to architectural factors rather than to hyperparameter asymmetries. Recent evidence shows that ResNets and Vision Transformers reach competitive accuracy when trained under comparable modern optimization schedules, supporting a unified configuration across architectures (Wightman et al., 2021). The chosen learning rate falls within the typical range used when fine-tuning pretrained backbones on small datasets like UC-Merced (2,100 images), while the weight decay of 0.01 corresponds to the standard choice for fine-tuning pretrained backbones. Even though Vision Transformers can benefit from smaller learning rates and higher weight decays, using equal hyperparameters is crucial for a fair comparison between architectures.

Model description.

The first architecture evaluated was ResNet-50, from the ResNet family proposed by (He et al., 2016). Its central innovation is the residual learning framework: instead of learning a direct mapping H(x), the network learns the residual F(x) = H(x) − x, so the final output becomes F(x) + x. These skip connections allow gradients to flow directly through the network during backpropagation, solving the vanishing gradient problem that had prevented training of plain networks deeper than ~20 layers. For this work, the model was initialized with weights pretrained on ImageNet-1K (~1.28M images, 1,000 classes), yielding 23.5 million trainable parameters.

Training was conducted for a maximum of 25 epochs using a learning rate of 1e-4 and a weight decay of 1e-4. The conservative learning rate preserves the pretrained weights during fine-tuning, while the weight decay provides L2 regularization to mitigate overfitting given the small dataset size. Early stopping was monitored on validation loss, and the best model checkpoint was saved based on the highest validation F1 score.

The Vision Transformer (ViT) introduced by (Dosovitskiy et al., 2021) demonstrated that a pure Transformer architecture, originally designed for natural language processing (NLP), can be applied directly to sequences of image patches with minimal modifications, achieving competitive or superior performance to CNN-based models when pretrained at sufficient scale. Unlike ResNet's convolutional approach, ViT processes images by dividing the input into a grid of non-overlapping 16×16 pixel patches — yielding 196 patches for a 224×224 image — which are flattened, linearly projected into a 768-dim embedding space, and treated as tokens. A learnable token is prepended to the sequence and learnable position embeddings are added to preserve spatial order, as the self-attention operation is inherently permutation-invariant.

The model was initialized with ImageNet-1K pretrained weights, yielding 85.8 million trainable parameters — approximately 3.7× more than ResNet-50. Training configuration and evaluation metrics were kept identical to ResNet-50 to ensure a fair architectural comparison, Early stopping was monitored on validation loss, and the best model checkpoint was saved based on the highest validation F1 score.

Results.

1) **General models performance:**

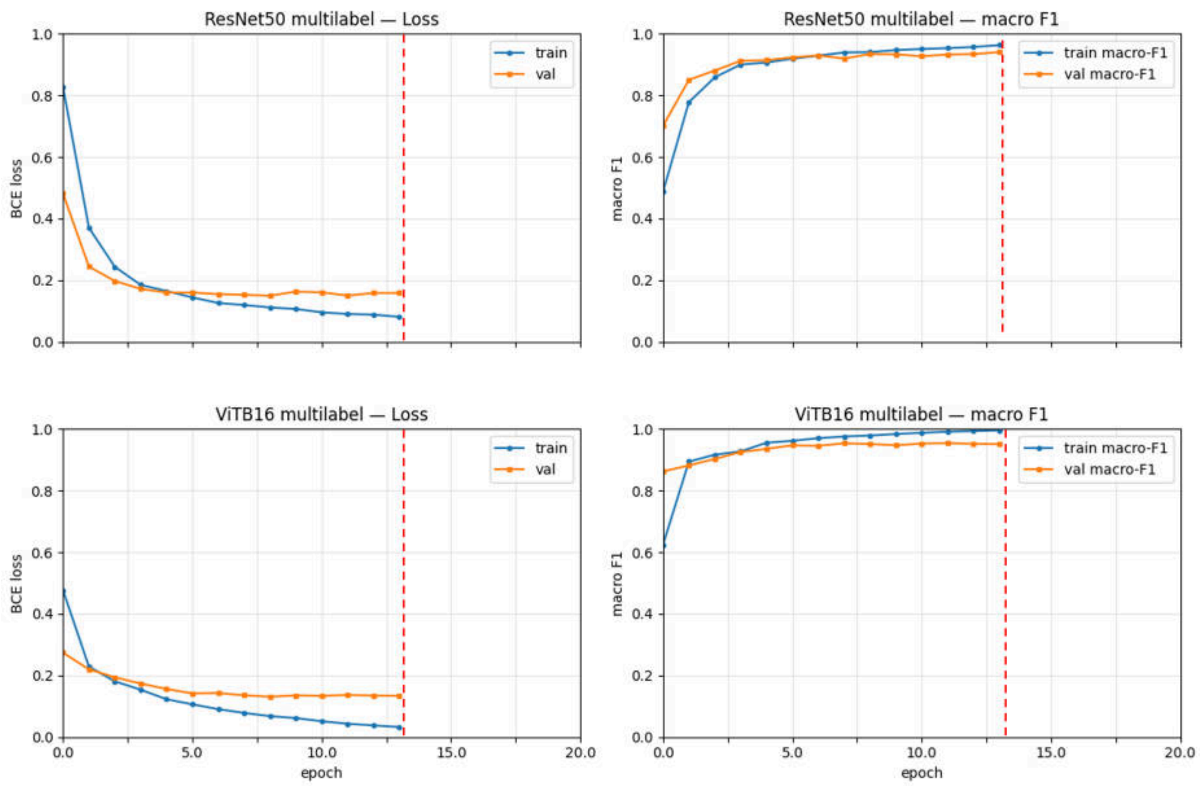


Figure 2: Model convergence during training. The dotted line corresponds to the early stopping epoch reached for both models. ResNet and ViT stopped at 13 epochs.

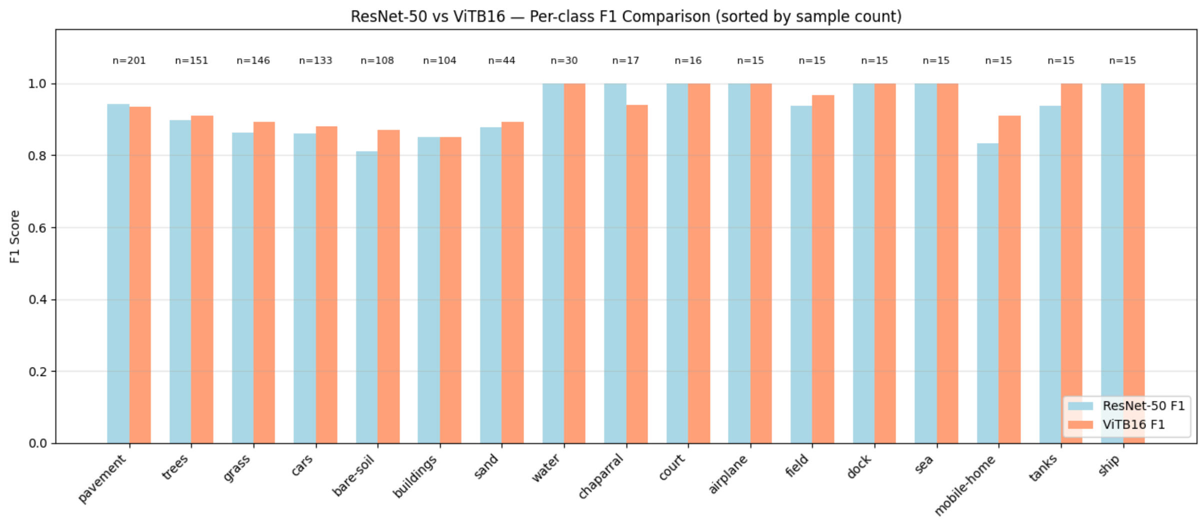
Figure 2 compares the convergence behavior of both models during training under the same optimization regime, monitored through Binary Cross-Entropy (BCE) loss and macro F1-score on the training and validation sets. Early stopping was configured with a minimum improvement of 0.001 on the validation loss and a patience of 5 epochs, and the final checkpoint was selected at the epoch that maximized the validation macro F1-score. Under these conditions, both models triggered early stopping at epoch 13 (red dashed line). ResNet-50 exhibits smooth and nearly parallel convergence of training and validation losses, which plateau at approximately 0.08 and 0.16 respectively, with training and validation macro F1 also tracking closely and stabilizing around 0.95. Despite this, ViT-B/16 achieves a lower validation loss and a comparable validation macro F1 to ResNet-50, confirming that overfitting on the training set does not translate into degraded generalization at the selected checkpoint.

Table 1 summarizes the performance of both pretrained models across all evaluation metrics. ViT-B/16 outperforms ResNet-50 consistently across every metric, though the margin varies by measure. The highest difference was observed on the Exact match accuracy metric (the most restricted one) where ViT outperformed ResNet by 0.06. A similar case was observed for the Macro F1 (0.94 vs 0.92) for ViT and ResNet respectively. Both models achieve high macro mAP (0.98 vs. 0.97), implying strong ranking quality across all 17 classes. Hamming loss confirms ViT's advantage at the individual label level (0.036 vs. 0.043), meaning roughly one percentage point fewer label-level errors per image. Despite the big difference between trainable parameters (85.8M vs. 23.5M), the use of kernel GPU/G4 made the training time negligible between models.

Performance Metric	ResNet-50	ViT-B/16
Accuracy	0.9562	0.9637
Macro F1	0.9266	0.9447
Micro F1	0.8933	0.9105
Samples F1	0.9034	0.9163
Macro mAP	0.9786	0.9802
Hamming Loss	0.0438	0.0363
Exact Match Accuracy	0.5096	0.5605

Table 1: Model comparison.

2) Individual class performance



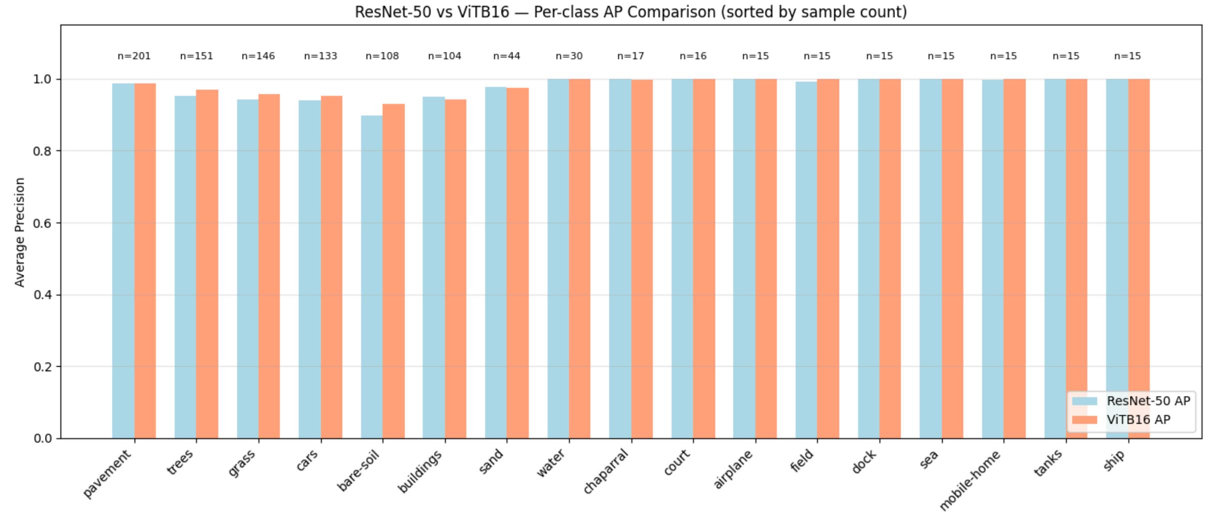
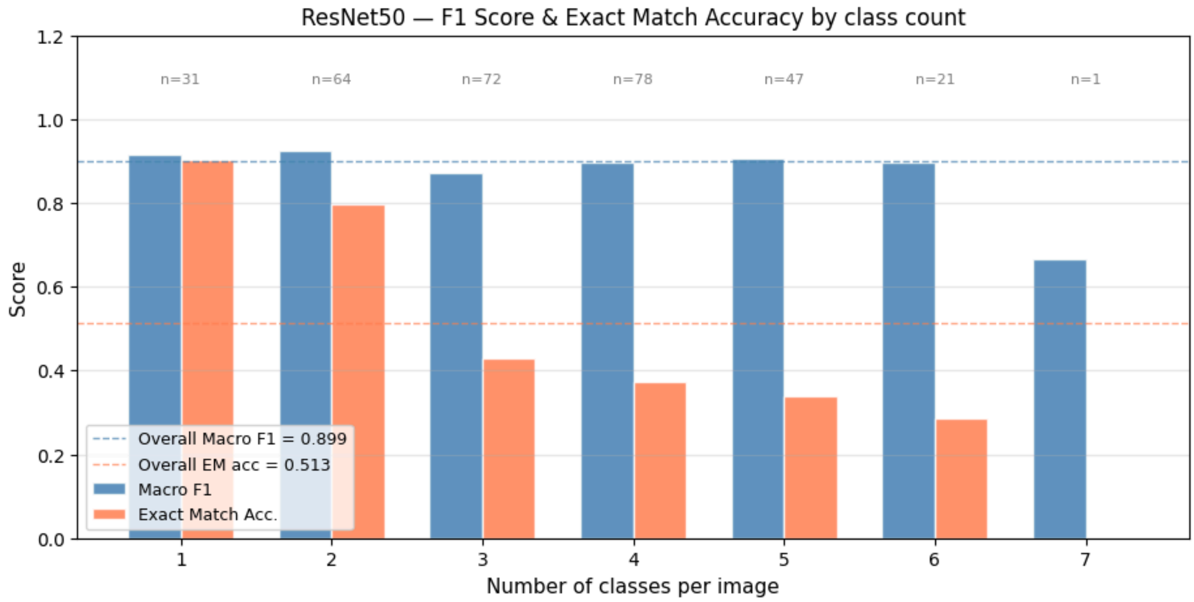


Figure 3: Model performance at class level considering F1 and mAP metrics

Figure 3 presents per-class F1 and Average Precision for both models. Contrary to what might be expected, ViT-B/16 does not show a consistent advantage specifically tied to class size — both models perform well on frequent classes such as *pavement* (n=201) and *trees* (n=151), where F1 scores exceed 0.90 for both architectures. The more meaningful pattern is that ViT-B/16 shows more consistent performance across the full class spectrum with higher F1. The most notable gap appears in: *mobile-home* (n=15), *fields* (n=15) and bare soil (n=108) where ResNet-50 drops noticeably while ViT-B/16 maintains higher F1 scores, suggesting that global self-attention better captures discriminative features when training signal is scarce (small training set).

The per-class AP comparison showed that both models performed well for all classes, achieving near-perfect AP on high-frequency classes. Taken together, both plots suggest ViT-B/16's advantage over ResNet-50 considering both metrics.

3) Individual class per image performance



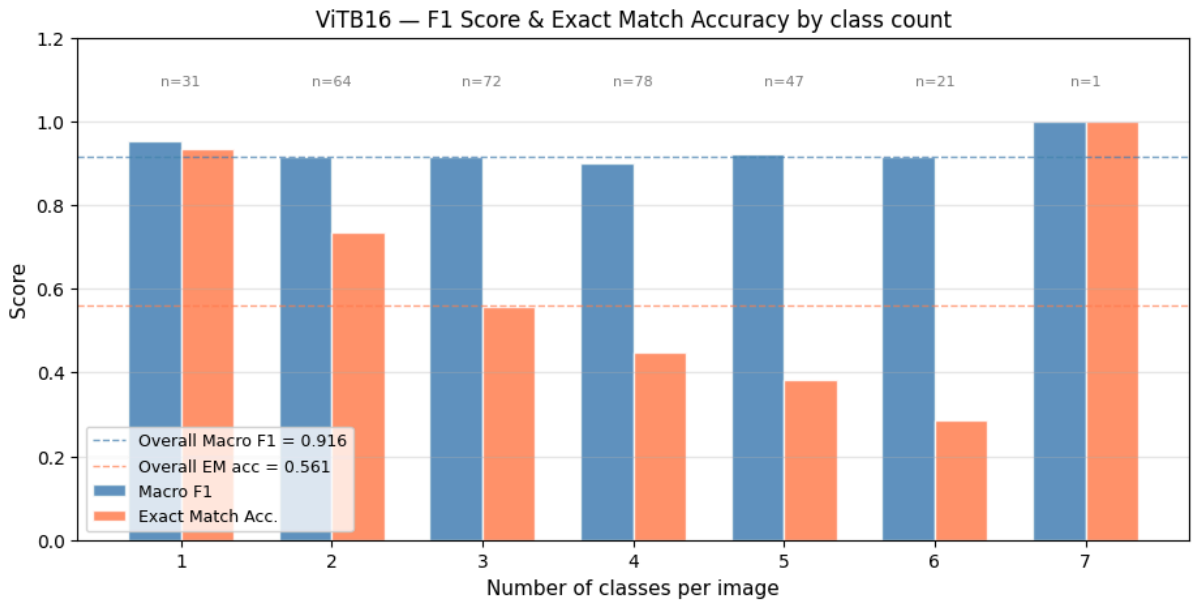


Figure 4: Model performance at number of classes per image level.

Figure 4 reveals a monotonic decline in Exact Match Accuracy for both models as the number of simultaneously active labels per image (k) increases. In both cases, the score drops from approximately 0.92 at k=1 to around 0.28 at k=6, since Exact Match requires the model to predict every single label correctly at once — meaning that even at 95% per-label accuracy, a single mislabelled class is enough to drop the sample score to 0. In contrast, the macro F1-score remains notably stable across all bins (0.87–0.92 for ResNet-50 and 0.89–0.95 for ViT-B/16), indicating that the per-label quality of the predictions does not deteriorate with scene complexity.. ViT-B/16 consistently outperforms ResNet-50 on Exact Match Accuracy throughout the medium-complexity range (k=3 to k=5), which concentrates most of the test set (197 of 314 images) and largely explains why its global EM (0.561) exceeds that of ResNet-50 (0.513) by a wider margin than the gap observed in macro F1 (0.916 vs. 0.899). The k=7 bin contains a single sample and should therefore be interpreted as anecdotal: ViT-B/16 recovers it perfectly while ResNet-50 fails entirely, but no statistically meaningful conclusion can be drawn from a single image.

Discussion/conclusion

<u>Why do ResNet50 and ViT perform similarly?</u>

Both ResNet-50 and ViT-B/16 confirmed the hypothesis that pre-trained models can handle the class and image imbalance of UC-Merced, achieving strong overall and per-class performance. ViT-B/16 outperformed ResNet-50 consistently across all metrics, with the largest gaps in Exact Match Accuracy (0.5605 vs. 0.5096) and Macro F1 (0.9447 vs. 0.9266), while both models achieved near-identical macro mAP (~0.98), indicating equally strong ranking quality across all 17 classes. Both triggered early stopping at epoch 13 under the same optimization regime. However, the training dynamics reveal an important trade-off: ResNet-50 exhibits a narrow training–validation gap indicative of good generalization and limited overfitting (Figure 2), whereas ViT-B/16 reaches a lower absolute training loss (~0.05) but stabilizes at a validation loss of ~0.13, producing a proportionally wider gap. This divergence is equally visible in the macro F1 curves, where ViT-B/16 training F1 approaches 1.0 while validation plateaus near 0.94. The larger relative gap suggests that ViT-B/16 is more prone to overfitting on UC-Merced, a behavior consistent with the weaker inductive biases of Transformer architectures on limited datasets (Steiner et al., 2022). Because the general differences between models is marginal, we cannot conclude that one model outperforms the other one for this particular data set and experiment. This strong performance by both architectures can be largely attributed to transfer learning: both models were pre-trained on millions of images, allowing their hidden layers to develop robust low-level feature detectors — such as edges, textures, and color gradients — that transfer effectively even to the relatively small UC-Merced dataset of 2,100 images. Given this constraint, the marginal performance gap between models is unsurprising. It is plausible, however, that on a substantially larger and more complex dataset — one with greater intra-class shape variability and finer-grained semantic distinctions — ViT-B/16 could more decisively outperform ResNet-50, as its self-attention mechanism enables the modeling of long-range spatial dependencies that convolutional architectures are inherently limited in capturing.

Why do these models perform well even if there are unbalanced classes?

Even though ViT performs slightly better in all cases, both models correctly classify with a high level of precision, even with classes that are under-represented or less trained on. This occurs for a number of reasons: 1) Pretrained models store edges, textures, shapes, part-whole hierarchies from ~1.28M natural images; and 2) Using posterior weight information in the BCEWithLogitsLoss function ensures that rare classes contribute proportionally more to the parameter updates. By stratifying the training, validation and test data according to these proportions we mitigate the tendency of multi-label networks to default to predicting majority labels.

A very important semantical conclusion needs to be made regarding the class description. The most underrepresented samples (chaparral, court, airplane, dock, sea, mobile home, tanks, ship) are objects that are easier to distinguish and confuse in an aerial image compared to more represented classes (trees, pavement, grass, cars, bare-soil). Therefore, our conclusion that the model performs well in class imbalanced datasets holds only for the UC Merced data specifically.

Why do these models perform poorly at higher labels per image?

Here we can see a gap between metric driven results and real world applications, even though the Macro F1 scores remain high, as k increases, the Exact Match decreases. This is primarily because the probability of missing 1 class increases the more classes there are in an image. Additionally, the representations trained using the ImageNet data are meant to classify a single image, this could lead to the models learning to "focus" on a single object within the image, meaning it could have a harder time "focusing" or learning multiple objects within the image. Since the Macro F1 score in insensitive to complexity, meaning missclassifying 1 class in an image with 2 classes yields a lower score than in an image with 6 classes, we can attribute the high Macro F1 scores at k=1=2 as proof that the models perform significantly better at single or 2-class classification.

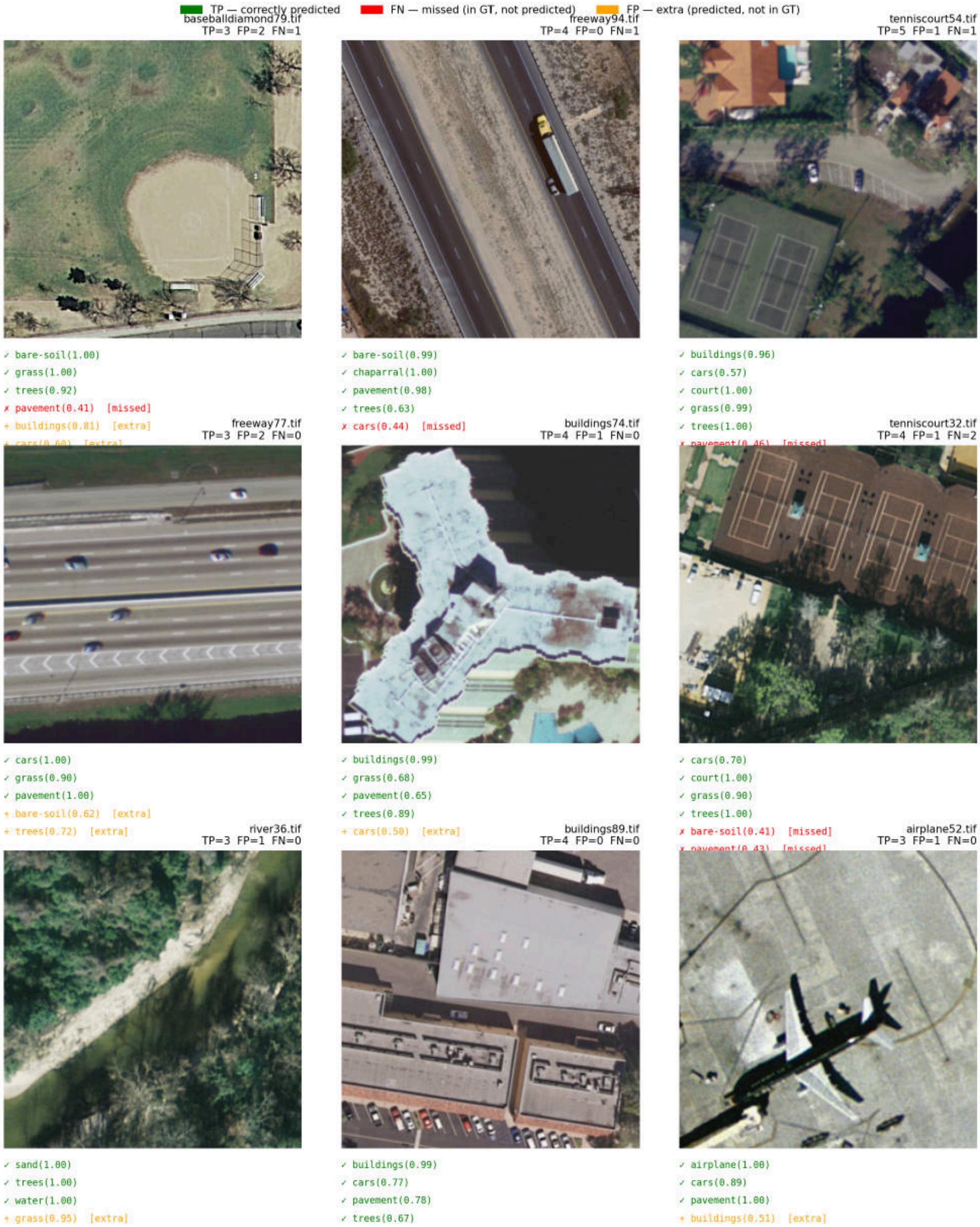
Limitations of work

Several limitations should be acknowledged. First, the absence of a fully balanced version of the UC-Merced dataset prevents a direct comparison between balanced and imbalanced training conditions, making it difficult to isolate the effect of class and image imbalance on model performance. Second, a fixed decision threshold of 0.5 could be applied uniformly across all classes; threshold optimization on a per-class basis could improve precision-recall trade-offs, particularly for underrepresented classes. Finally, computational constraints restricted the scope of hyperparameter tuning, meaning that the reported results may not reflect the optimal configuration for either architecture — a more exhaustive search over learning rate schedules, regularization strengths, and fine-tuning strategies could yield further performance gains.

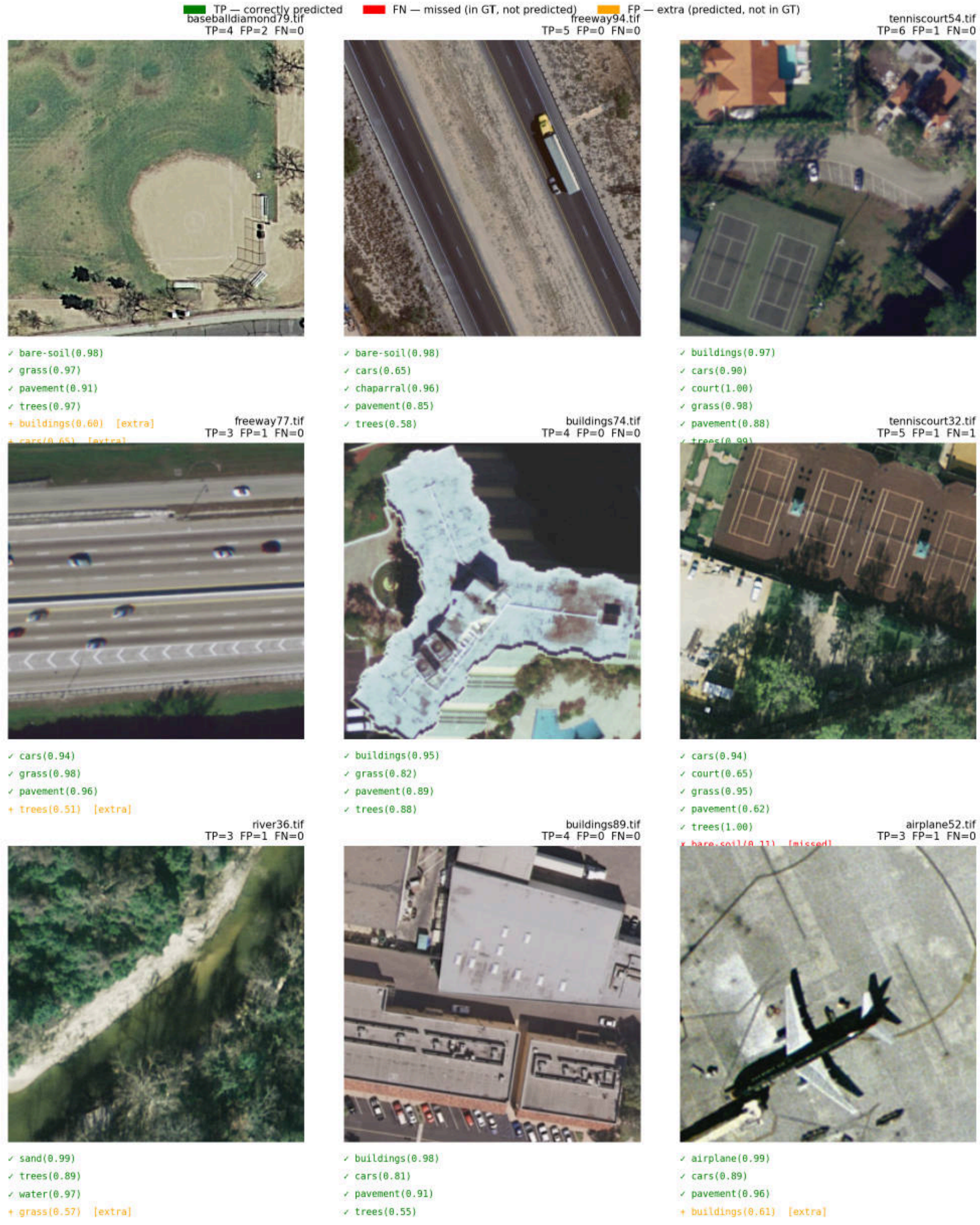
Future recommendations:

Future work should explore more recent architectures such as Swin Transformer, ConvNeXt, and hybrid CNN-Transformer models, which may better balance spatial inductive biases and global context modeling. Evaluating the framework on larger remote sensing datasets — such as AID, NWPU-RESISC45, or BigEarthNet — would provide a more rigorous assessment of scalability and allow ViT-B/16's self-attention mechanism to demonstrate its full potential. On the training side, stronger regularization strategies (e.g., MixUp, CutMix, stochastic depth) and asymmetric loss functions designed for multilabel imbalance could mitigate the overfitting observed in ViT-B/16. Self-supervised pre-training on domain-specific remote sensing data using frameworks such as MAE or DINO represents a promising alternative to ImageNet initialization.

Appendix.



A1: Image examples for pretrained ResNet50 model



A2: Images examples for pretrained ViT Base 16

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AI Statement

- Claude code IA model opus 4.7 was used on this project for code structure considering the implementation of the utils.py. 00_Resnet and 00_ViT files were created considering claude recommendations on checkpoint and early stopping sections.
- Claude IA was also used as reference searcher for this project were all the DOIs from the paper were required for the correct check, avoiding IA hallucination (fake citation and invented bibliography)
- The main core of the report corresponds to information from the lectures, papers and practical code. Claude IA was used for grammatical semantics and coherence of the sentences. In specific sections on the methods, results and conclusions, more intense prompt over interpretation of the architecture, hyperparameters and potential limitations of the project were used.
- README.md file was also created based on a summarise prompt and check before pushing it into the git.

Source Code: https://github.com/gabrielcastrob/Deep_learning_WUR.git